Extended Case 6 — Harnswell Quality

Precision parts and process stability · 307312 Applied Statistics — Masters Level

- **Start after:** descriptive statistics and inference in Parts 2 and 3.
- **Optional advanced case:** introduces statistical process control; read the chart primer below first.
- **Suggested time:** 90–120 minutes.
- **Business question:** Is the cam-roller grinding process stable, and does its output meet engineering specifications?

Learning outcomes:

- Distinguish engineering specifications from statistical control limits.
- Preserve rational subgroups when constructing X-bar and R charts.
- Explain why process stability must precede capability claims.

1. The quality manager's brief

- Harnswell manufactures precision cam rollers for sewing machines.
- Oversized parts may be downgraded; undersized parts may become scrap.
- **Source:** *Basic Business Statistics*, ninth edition, Chapter 18, pp. 752–754.
- **Unit:** one roller diameter, in inches; five measurements per production batch for 30 batches.
- **Engineering limits: 0.5072 to 0.5075 inches**, inclusive.
- **Period:** source batch order is retained; calendar dates are not supplied in the CSV.
- The workbook column `Day` is represented here as `batch`, consistent with the case's production subgroups.
- **Limit:** the historical case narrative does not establish that temperature alone caused the observed variation.

2. Import and preserve batch membership

```
source("lessons/_includes/course-tools.R")
rollers <- read.csv("data/business/extended-harnswell.csv")
stopifnot(nrow(rollers) == 150, all(table(rollers$batch) == 5))
LSL <- .5072
USL <- .5075
knitr::kable(head(rollers), digits = 4)
```

batch	diameter
1	0.5076
1	0.5076
1	0.5075
1	0.5077
1	0.5075
2	0.5075

- `data/business/extended-harnswell.csv`; guide: `data/business/README.md`.
- Never sort all diameters and then divide them into new batches; that destroys the subgroup structure.
- `LSL` and `USL` are engineering requirements, not limits estimated from these measurements.

3. Describe specification compliance

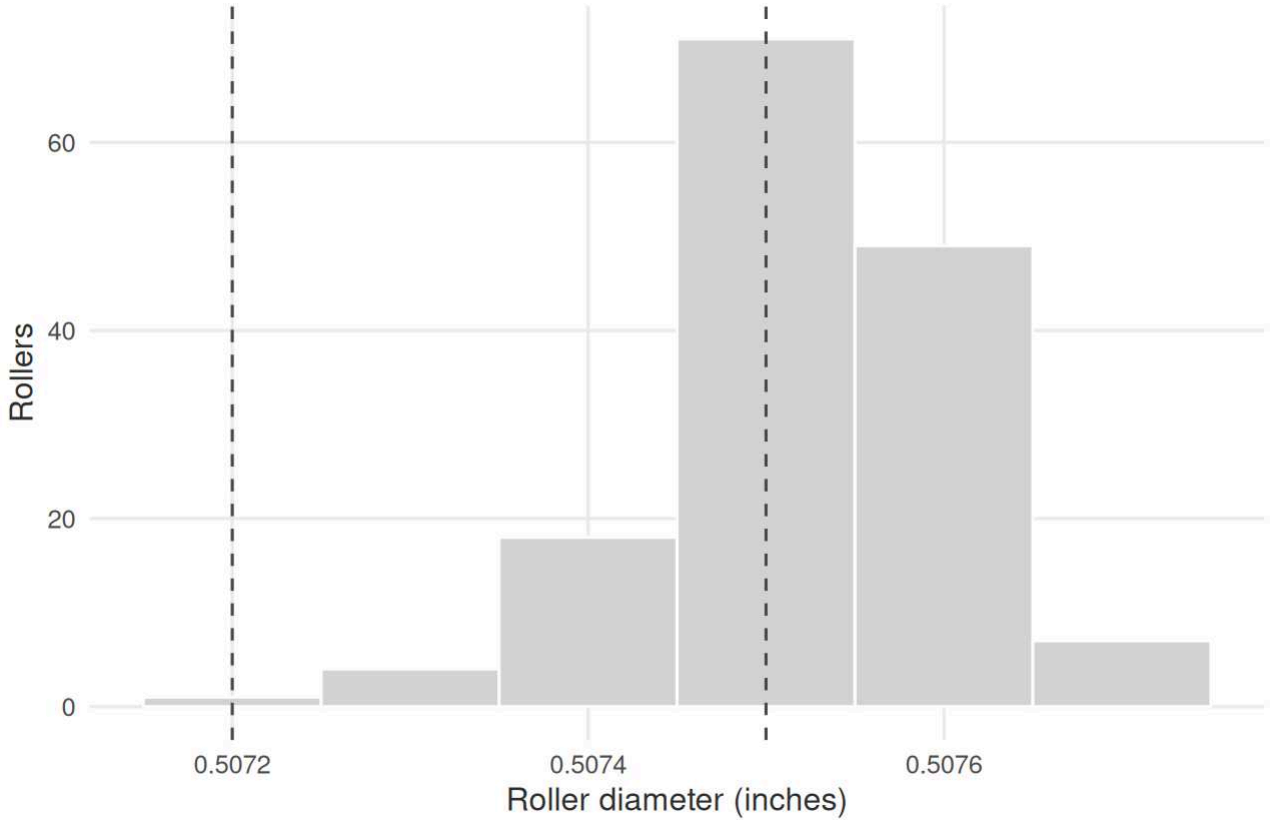
```
knitr::kable(rollers %>% summarise(n = n(), mean = mean(diameter),
  sd = sd(diameter), min = min(diameter), max = max(diameter)), digits = 6)
```

n	mean	sd	min	max
150	0.507523	8.7e-05	0.5072	0.5077

```
compliance <- rollers %>% mutate(status = case_when(
  diameter < LSL ~ "Below specification",
  diameter > USL ~ "Above specification",
  TRUE ~ "Within specification")) %>%
  count(status) %>% mutate(percent = 100 * n / sum(n))
knitr::kable(compliance, digits = 2)
```

status	n	percent
Above specification	56	37.33
Within specification	94	62.67

```
ggplot(rollers, aes(diameter)) +
  geom_histogram(binwidth = .0001, boundary = .50715,
                 fill = petra_accent, colour = "white") +
  geom_vline(xintercept = c(LSL, USL), colour = petra_ink, linetype = "dashed") +
  scale_x_continuous(labels = function(x) sprintf("%.4f", x)) +
  labs(x = "Roller diameter (inches)", y = "Rollers")
```



- The workbook has **56 above-specification rollers (37.33%)**, 94 within specification and none below.
- The book narrative reports 55 above specification; the table here follows the unchanged workbook values.
- Do not extrapolate this observed percentage to annual production without a defensible sampling and stability basis.

4. A short process-control primer

- Within-batch variation concerns the five rollers in one subgroup.
- Between-batch changes may indicate shifts in centering or operating conditions.
- An **R chart** tracks within-batch ranges; an **X-bar chart** tracks batch means.
- Check the R chart first: unstable within-batch variation undermines interpretation of X-bar limits.
- Control limits estimate expected variation for a stable process; specification limits describe acceptable products.
- A stable process can still produce unacceptable parts.

5. Estimate initial chart limits

- For subgroup size five, use **A2 = 0.577, D3 = 0, D4 = 2.114**.
- X-bar limits: grand mean ± A2 × average range.
- R limits: D3 × average range and D4 × average range.
- These are initial Phase I limits estimated using all 30 batches, including any unusual batches.

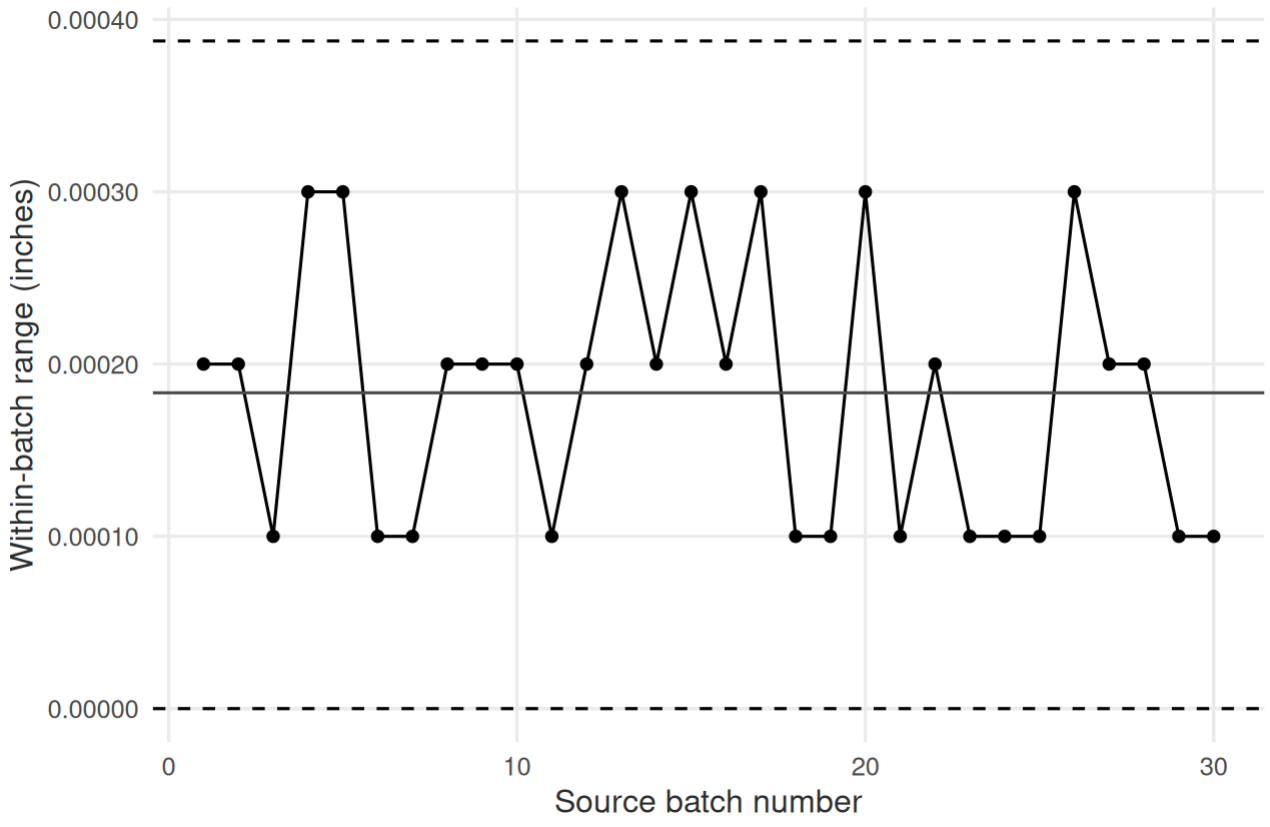
```r
subgroups <- rollers %>% group_by(batch) %>%
  summarise(xbar = mean(diameter), R = max(diameter) - min(diameter),
    n = n(), .groups = "drop") %>% arrange(batch)
stopifnot(all(subgroups$n == 5))
xbarbar <- mean(subgroups$xbar)
Rbar <- mean(subgroups$R)
xlimits <- xbarbar + c(-1, 1) * .577 * Rbar
rlimits <- c(0, 2.114 * Rbar)
knitr::kable(data.frame(chart = c("X-bar", "R"),
  centre = c(xbarbar, Rbar), lower = c(xlimits[1], rlimits[1]),
  upper = c(xlimits[2], rlimits[2])), digits = 7)
```

chart	centre	lower	upper
X-bar	0.5075227	0.5074169	0.5076284
R	0.0001833	0.0000000	0.0003876

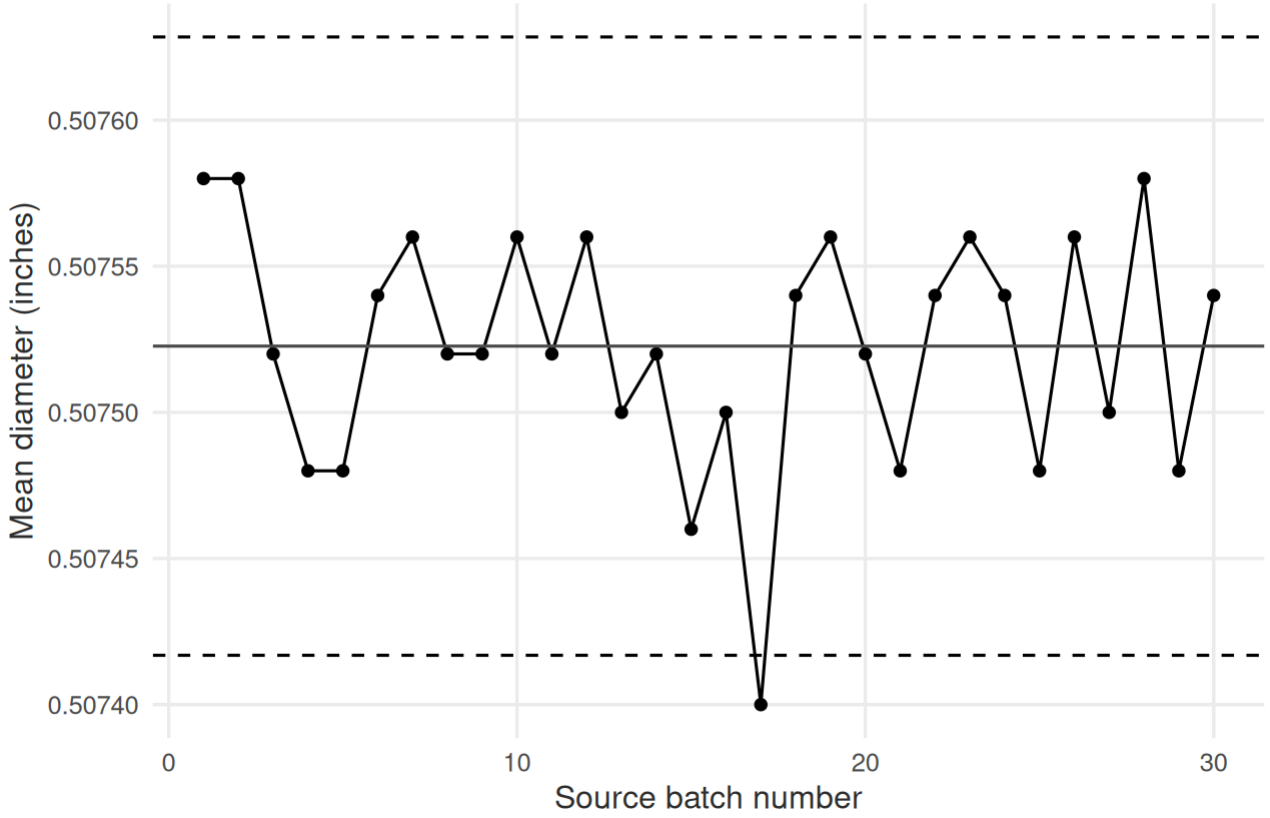
- `group_by(batch)` preserves the original subgroups.
- The X-bar limits apply to **batch means**, not individual roller diameters.
- Engineering limits should not replace statistical limits on these charts.

6. Check the R chart, then the X-bar chart

```r
ggplot(subgroups, aes(batch, R)) + geom_line() + geom_point() +
  geom_hline(yintercept = Rbar, colour = petra_ink) +
  geom_hline(yintercept = rlimits, linetype = "dashed") +
  scale_y_continuous(labels = function(x) sprintf("%.5f", x)) +
  labs(x = "Source batch number", y = "Within-batch range (inches)")
```



```r
ggplot(subgroups, aes(batch, xbar)) + geom_line() + geom_point() +
  geom_hline(yintercept = xbarbar, colour = petra_ink) +
  geom_hline(yintercept = xlimits, linetype = "dashed") +
  scale_y_continuous(labels = function(x) sprintf("%.5f", x)) +
  labs(x = "Source batch number", y = "Mean diameter (inches)")
```




```r
flags <- subgroups %>% mutate(xbar_out = xbar < xlimits[1] | xbar > xlimits[2],
  R_out = R < rlimits[1] | R > rlimits[2]) %>% filter(xbar_out | R_out)
knitr::kable(flags, digits = 6)
```

batch	xbar	R	n	xbar_out	R_out
17	0.5074	3e-04	5	TRUE	FALSE

- The initial X-bar chart flags **batch 17** below its lower limit; the R chart has no beyond-limit point.
- A signal is a reason to investigate a potential special cause.
- No beyond-limit points would not by itself prove stability: runs, trends, grouping and operational context also matter.
- Keep batch 17 in the original analysis; do not delete it merely to make the chart look stable.

7. Capability requires a stable, suitable process model

- **Cp** compares specification width with six within-process standard deviations.
- **Cpk** also accounts for the process mean's distance from each specification limit.
- With subgroup size five, an estimate of within-process SD is average range / **d2**, where **d2 = 2.326**.
- The initial charts signal instability, so a reliable capability claim is not supported by this initial dataset.
- The following template is intentionally not executed. Use it only after a documented stability investigation and suitable distribution/measurement checks.

```r
# stable_subgroups must come from a justified, documented process phase.
mu <- mean(stable_subgroups$xbar)
sigma_within <- mean(stable_subgroups$R) / 2.326
Cp <- (USL - LSL) / (6 * sigma_within)
Cpk <- min((USL - mu) / (3 * sigma_within),
           (mu - LSL) / (3 * sigma_within))
```

- A negative Cpk indicates a fitted mean beyond a specification limit; it is not a negative defect percentage.
- Do not convert Cp or Cpk into a guaranteed future defect rate without the additional model assumptions.

8. Independent quality-improvement briefing

1. Reproduce specification counts and percentages from the 150 measurements.
2. Report the initial R and X-bar limits with units and appropriate decimal precision.
3. Explain how batch 17 can signal a shift even though none of its individual measurements is below specification.
4. Propose records to examine: temperature, warm-up, setup, tooling and measurement calibration.
5. The book describes a cold-start explanation and corrective action. Explain why the narrative is insufficient to prove causality in the CSV alone.
6. As a **retrospective sensitivity analysis**, recalculate limits excluding batch 17, retain the original results and state what evidence would justify treating it as a separate process condition.
7. Explain why recalculated limits alone do not validate future process stability or capability.
8. Recommend how to collect new subgroups after an intervention, and distinguish reducing variation from shifting the process mean.

Submission

- Specification table, both initial control charts and the flagged-batch table.
- A 150–200-word improvement memo addressing stability, compliance, the source discrepancy and next measurements.
- Include a justified comparison of original and sensitivity results if assigned; keep capability claims conditional.